1. Data Processing

1.1 Data structure and reshaping

We start with 100 synthetic ETFs data arrives in **long** format (one row per ETF per day). We reshaped the data to **wide** format which enables **vectorised operations**: instead of looping over each ETF, we compute across all 100 simultaneously using matrix arithmetic (e.g. a cumsum-based rolling mean), which gives roughly a 100× speed-up over a naive nested loop.

1.4 Time Series Split and Cross-Validation Setup

Following the Week 8 tutorial we split data into training (2010-2012) and test (2013+) periods, and define rolling CV folds within training. We also define the two helper functions used throughout Section 3: `flatten_features` converts the 3D feature array to the 2D matrix LightGBM expects, and `compute_ic` measures the Information Coefficient (mean daily Spearman correlation of predicted vs actual returns).

The CV scheme is illustrated conceptually as a sliding window: each fold's training window is 252 days, the validation window is 21 days, and a gap of exactly `lookahead` days separates them — guaranteeing no target labels from the training period contaminate the validation labels.

2. Feature Engineering

Our features extend the Week 8 tutorial's feature set. We compute all 40 base features for the entire training period at once using `compute_all_features` in our strategy.

2.1 Week 8 Baseline Features

We reuse the following technical indicators from the Week 8 tutorial: RSI (14-day relative strength), Bollinger Bands (price position vs. 20-day SMA $\pm 2\sigma$), MACD and PPO (trend indicators from 12/26-day EMA spreads), multi-horizon returns over 1/5/10/21/42/63 days, cross-sectional deciles of each return horizon across the 100 ETFs (which strip out common market moves), and date/rank features (month, weekday, and fractional rank of log dollar volume).

NATR, **ATR**, **sector**, and **sector-quantile returns** are omitted, our dataset consists of synthetic ETFs with no sector classification.

2.2 Features additional to Week 8

Beyond the Week 8 set, we add: the **MACD histogram** MACD_hist = MACD − Signal with $\text{Signal}_t = \text{Signal}_{t-1} + \frac{2}{10}(\text{MACD}_t - \text{Signal}_{t-1})$, capturing momentum acceleration; **price-to-SMA spreads** $(\text{close} - \text{SMA}_K)/\text{close}$ for $K \in \{5, 10, 20, 50, 200\}$ and the **crossover signals** $(\text{SMA}_5 - \text{SMA}_{20})/\text{close}$ and $(\text{SMA}_{10} - \text{SMA}_{50})/\text{close}$, measuring distance from trend and short-vs-long term alignment; **realised volatility** $\text{rvol}_w = \sqrt{\widehat{\text{Var}}(\log(\text{cl}_t/\text{cl}_{t-1}))_{w\text{-day}} \times 252}$ for $w \in \{5, 10, 21, 63\}$; **volume features** $\text{vol_ratio} = \text{vol}/\text{SMA}_{20}(\text{vol})$, $\text{vol_trend} = (\text{SMA}_5(\text{vol}) - \text{SMA}_{20}(\text{vol}))/\text{SMA}_{20}(\text{vol})$, and $\text{log_dollar_vol} = \log(1 + \text{close} \times \text{vol})$; the **open-close return** $(\text{close} - \text{open})/\text{open}$; **z-scores** $(\text{close} - \text{SMA}_K)/\sigma_K$ for $K \in \{20, 50\}$; and **rate of change** $\text{roc}_k = r_k \times 100$ for $k \in \{5, 21\}$ — linearly redundant with r_5/r_{21} but rescaled, giving tree splits finer-grained thresholds in the small-return region.

2.3 Feature Summary

The final feature set contains 40 engineered features across eight categories: 6 technical indicators (RSI, Bollinger Bands, MACD, PPO, MACD histogram), 7 price/moving-average spreads and crossovers, 6 multi-horizon returns, 4 multi-window realised volatility measures, 4 volume-based features (ratio, trend, log dollar volume, and open-close return), 4 rate of change/z-score features, 2 calendar features (month and weekday), and 7 cross-sectional features (deciles plus dollar-volume rank).

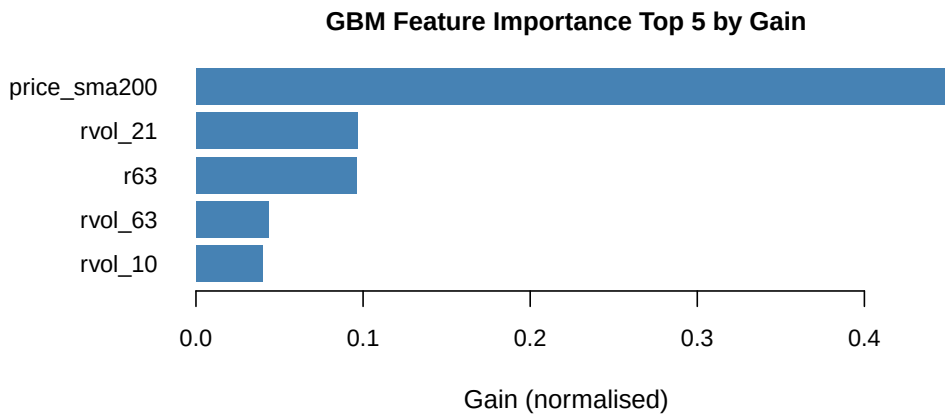
3. Model Development

Following the Week 8 tutorial we use **LightGBM** to predict cross-sectional ETF returns. We treat the **prediction horizon** (candidate values: 5, 10, 21 days) alongside model complexity parameters as joint hyperparameters, tuning all via time-series CV with **IC** (Information Coefficient) as the evaluation metric.

No look-ahead risk from precomputation. We precomputed `feat_3d` on the full training period before constructing CV folds. This is safe because each feature `feat_3d[i, j, k]` depends only on prices and volumes up to and including day i: RSI and EMA use only recursive past-value smoothing; rolling variances use only cumulative sums of past observations; cross-sectional deciles rank the 100 ETFs within day i only. We deliberately avoid global normalisation (e.g., dividing by the period-wide standard deviation), which would genuinely leak future information.

3.1 Fold Evaluation

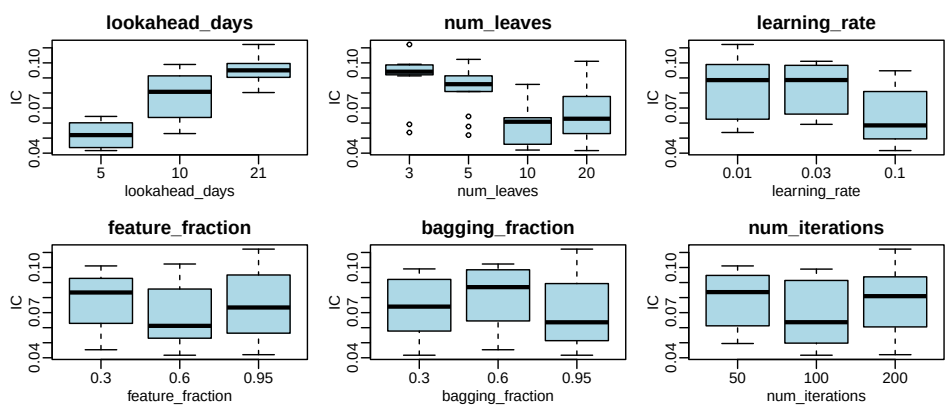
We first evaluate the single fold GBM with default hyperparameters, obtaining a mean IC of **0.156**. The feature-importance plot below identifies `price_sma200` as the dominant predictor.



Furthermore, evaluation across all available CV folds confirms a robust IC and stable predictive performance across different time periods.

3.2 Hyperparameter Tuning

We search jointly over **prediction horizon** (lookahead: 5, 10, 21 days) and model complexity parameters.



Selected hyperparameters (`lookahead = 21`, `num_leaves = 3`, `learning_rate = 0.03`, `bagging_fraction = 0.6`, `num_iterations = 200`): a 21-day horizon captures medium-term momentum, while shallow trees with a slow learning rate and strong regularisation prevent overfitting on noisy ETF returns. The mean IC improved from **0.156** (default) to **0.185** after tuning.

4. Final Strategy Explanation

4.1 Step-by-Step: How a Trade is Made

The complete trading pipeline runs inside `initialise_state` and `trading_algorithm` in `GroupA.R`. Below we trace each step explicitly using real data.

Step 1 — Initialise: Build matrices and train the LightGBM model.

Step 2 — Receive new data: append today's prices and drop the oldest row.

Step 3 — Extract features: call `extract_today_features` for all 100 ETFs at once.

Step 4 — Predict and Rank and generate trades: long top-10, short bottom-10.

Step 5 — Position rolling window: insert the new position and track the 21-layer stack.

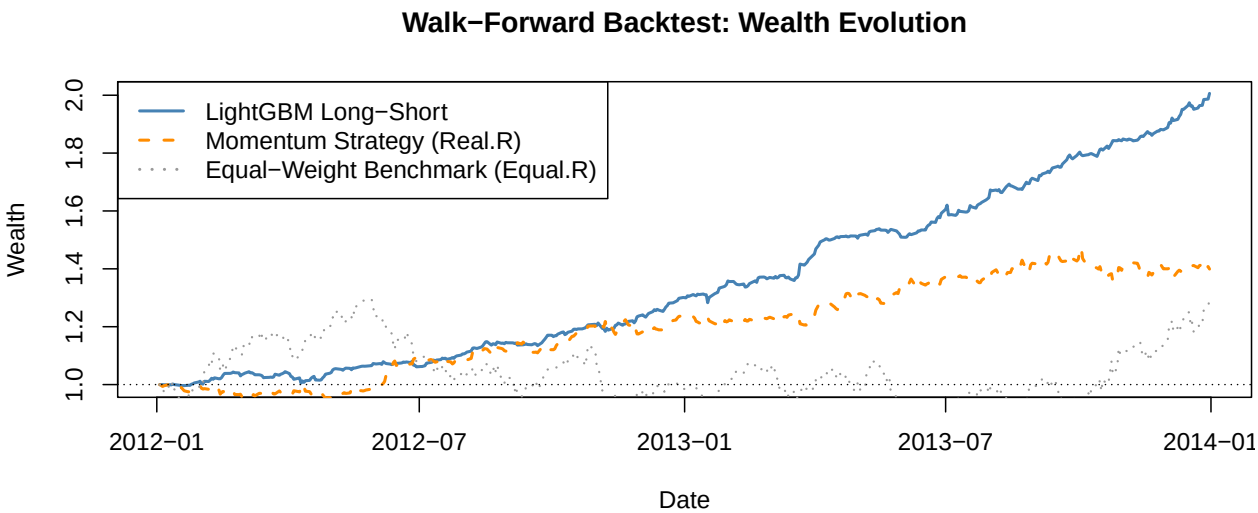
Position rolling window. The state holds a `[21 x 100]` positions matrix. Each day the new position vector is inserted at row 1 and the oldest row is dropped. Daily P&L is `sum(colSums(positions) * r1d)` — at steady state 21 overlapping layers are active, which is why `pos_size` is divided by `lookahead = 21`.

Incremental EMA/RSI updates (O(1)). Between retrains, `trading_algorithm` updates EMA12, EMA26, MACD signal, and RSI gain/loss averages with a single multiplication per symbol per indicator. Every 21 days a full batch recompute from `compute_all_features` resets the state to prevent floating-point drift.

Warmup period (200 days). SMA200 needs 200 days of history; `build_training_data` skips all rows with index ≤ 200, ensuring every training sample has valid features.

4.2 Performance Summary

We validate using the `walk_forward` function from `Test/walk_forward.R` (inlined to avoid its standalone execution block), mirroring the grader's evaluation exactly. The **equal-weight buy-and-hold** from `Equal.R` is the baseline.



The momentum benchmark ranks ETFs by a single past-return score $s_i^{\text{mom}}(t) = r_i^{\text{past}}(t)$, implicitly assuming $\mathbb{E}[r_{i,t+1} \mid \mathcal{F}_t]$ depends mainly on this one feature. LightGBM instead models the conditional mean $\hat{r}_{i,t+1} \approx \sum_{k=1}^{K} f_k(x_{i,t})$ on a p-dimensional feature vector, so it can capture non-linearities and interactions that a one-dimensional rule cannot. When the true signal is multi-dimensional, this reduces approximation error and can produce better cross-sectional rankings. Because all three strategies are evaluated under the same walk-forward procedure, transaction cost, and portfolio constraints, the comparison is fair: any performance difference comes from the signal-generation method rather than the backtest implementation.